

OrbWeaver: A Geometric Optics-Based RIS for Omnidirectional 3D Beam Steering

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[*] These authors contributed equally to this work.

Next-generation wireless applications demand high-capacity and low-latency connectivity



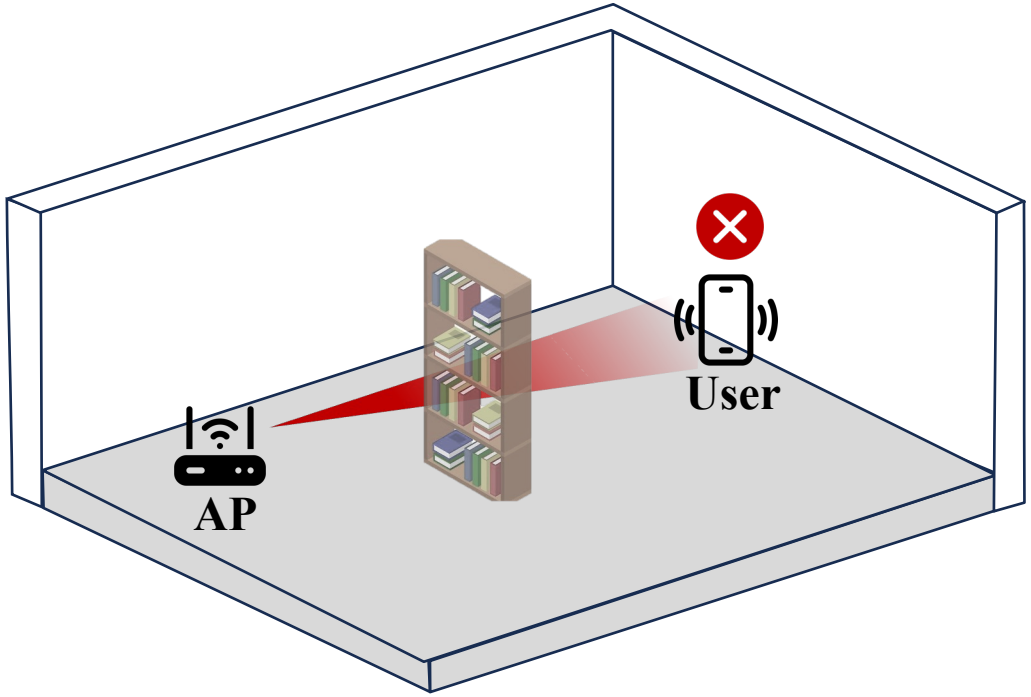




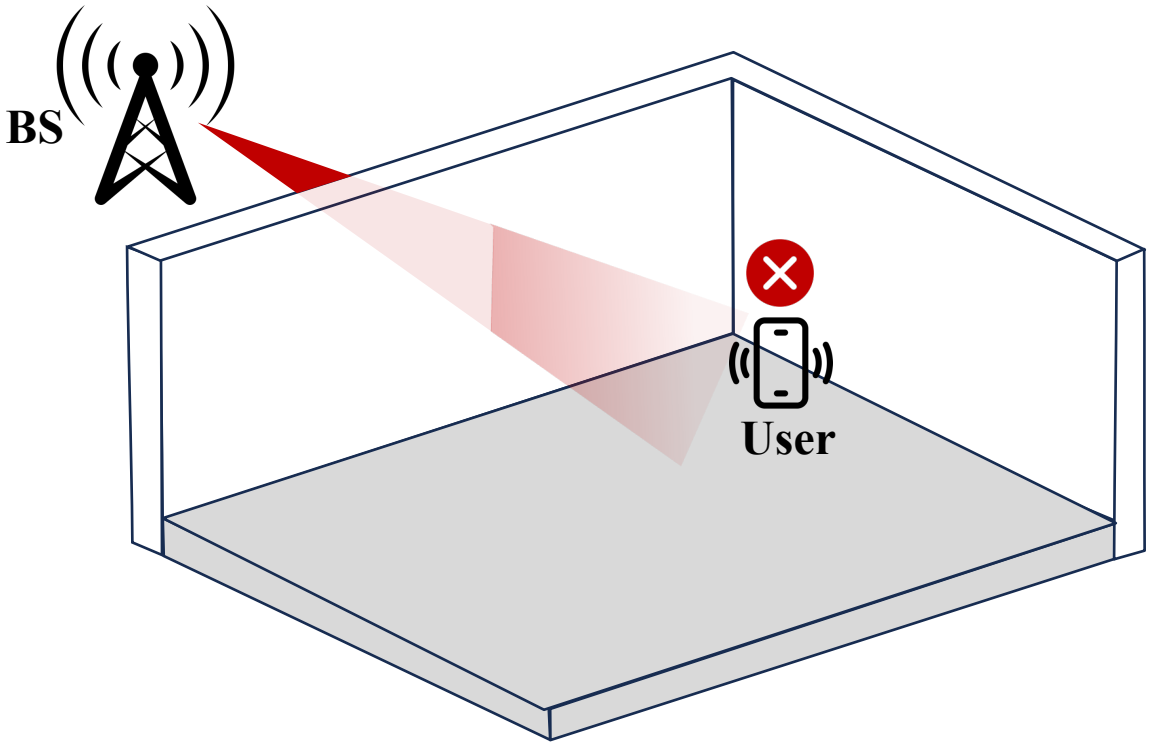


To support next-generation wireless applications • • •

However, high-frequency links are highly vulnerable to blockage

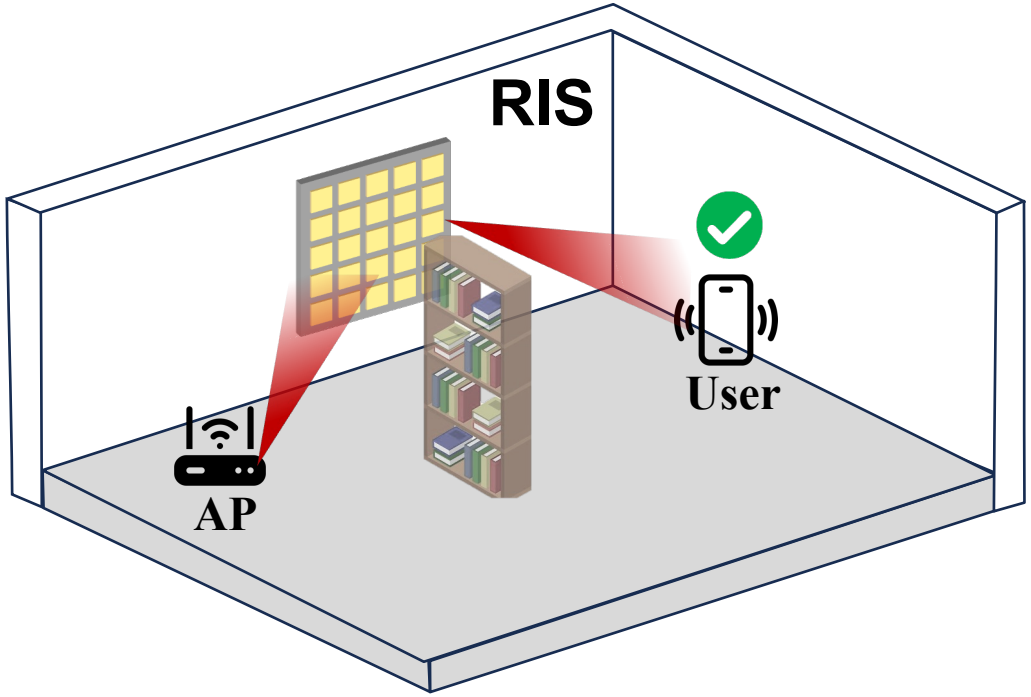


Obstacles Blockage

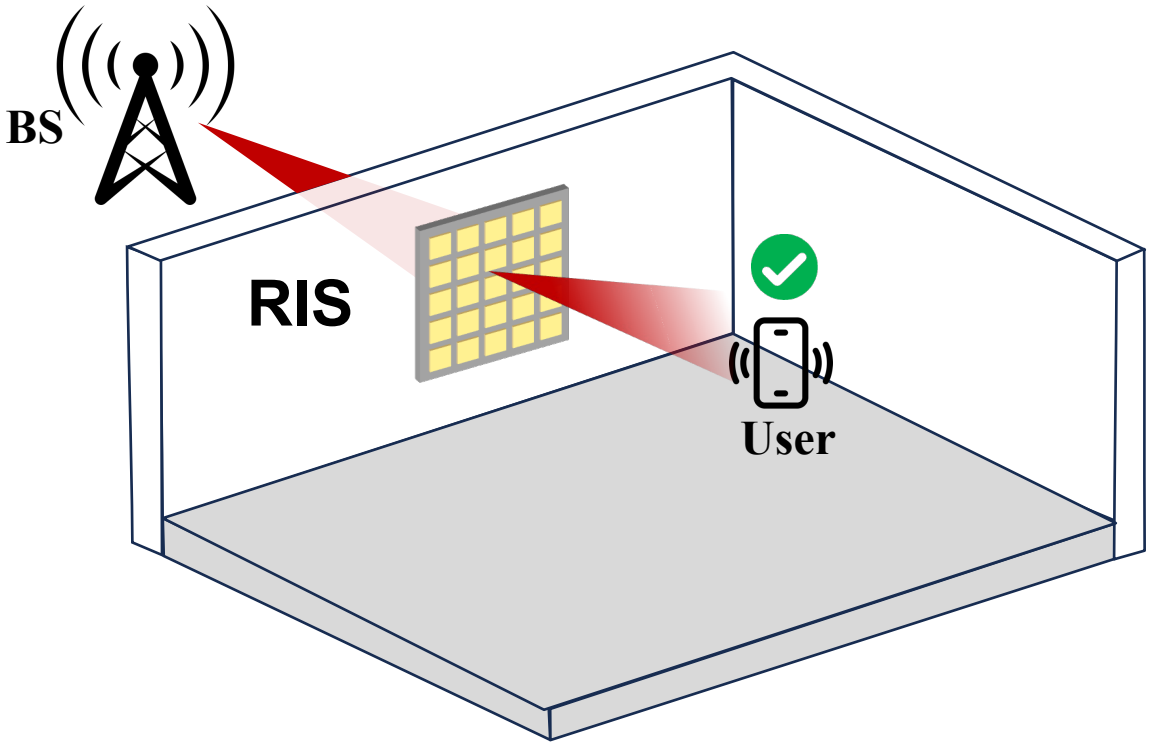


Wall Penetration

Reconfigurable intelligent surfaces reshape the wireless environment

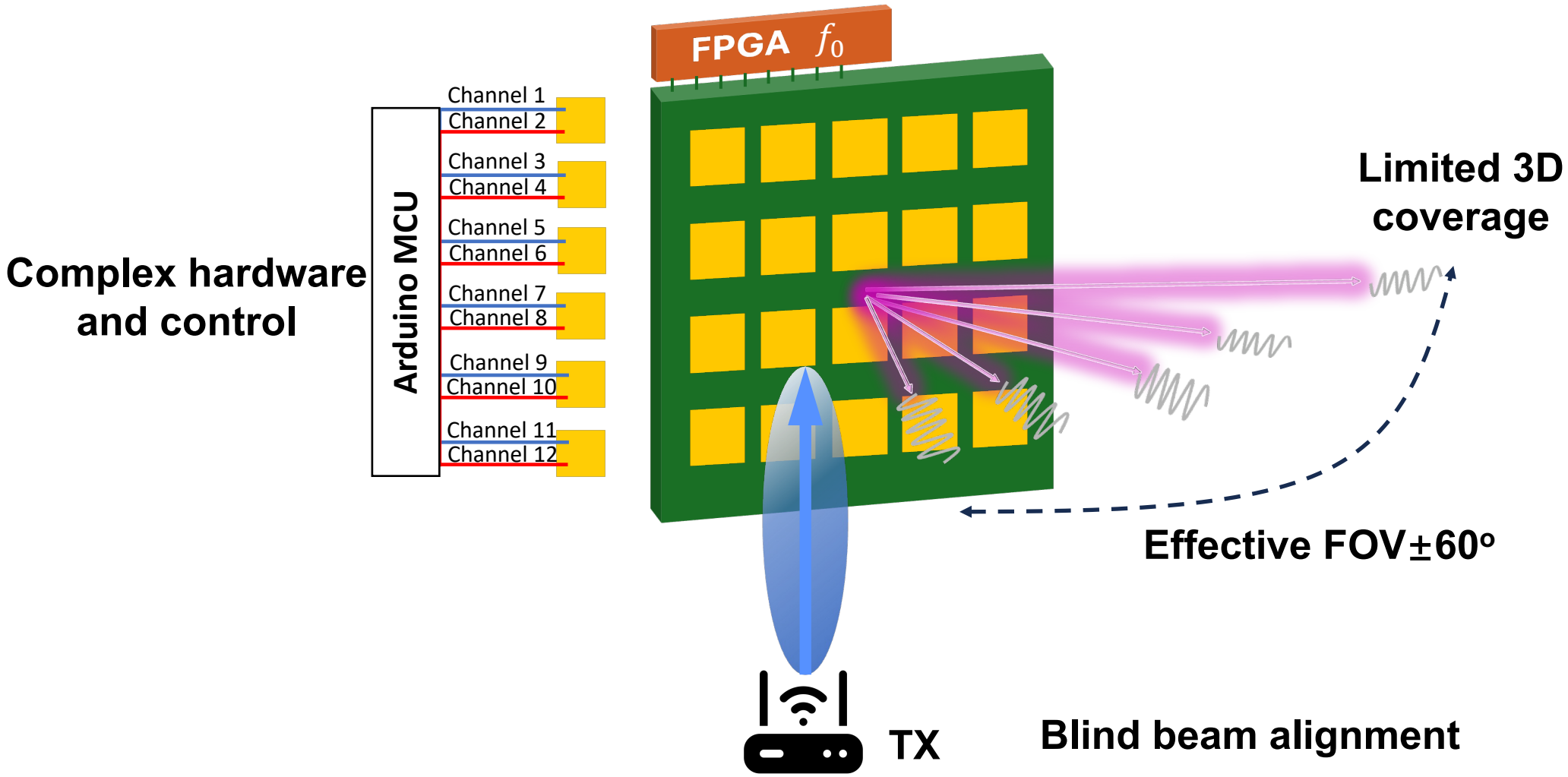


Non-specular reflection for routing around obstacles



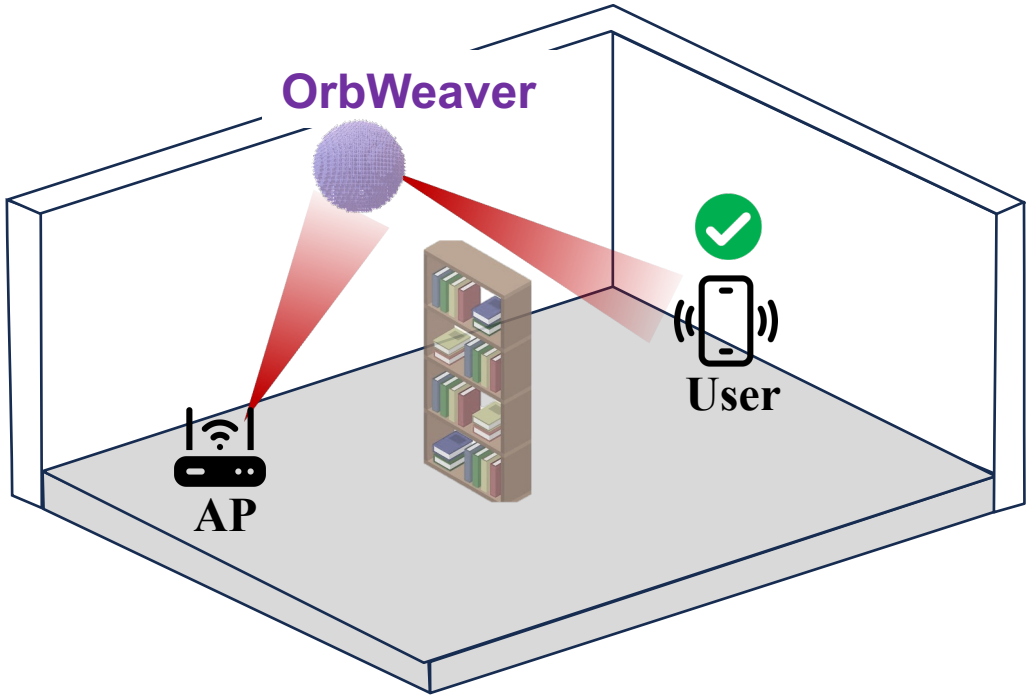
Bring outdoor signal indoors

Reconfigurable intelligent surfaces reshape the wireless environment

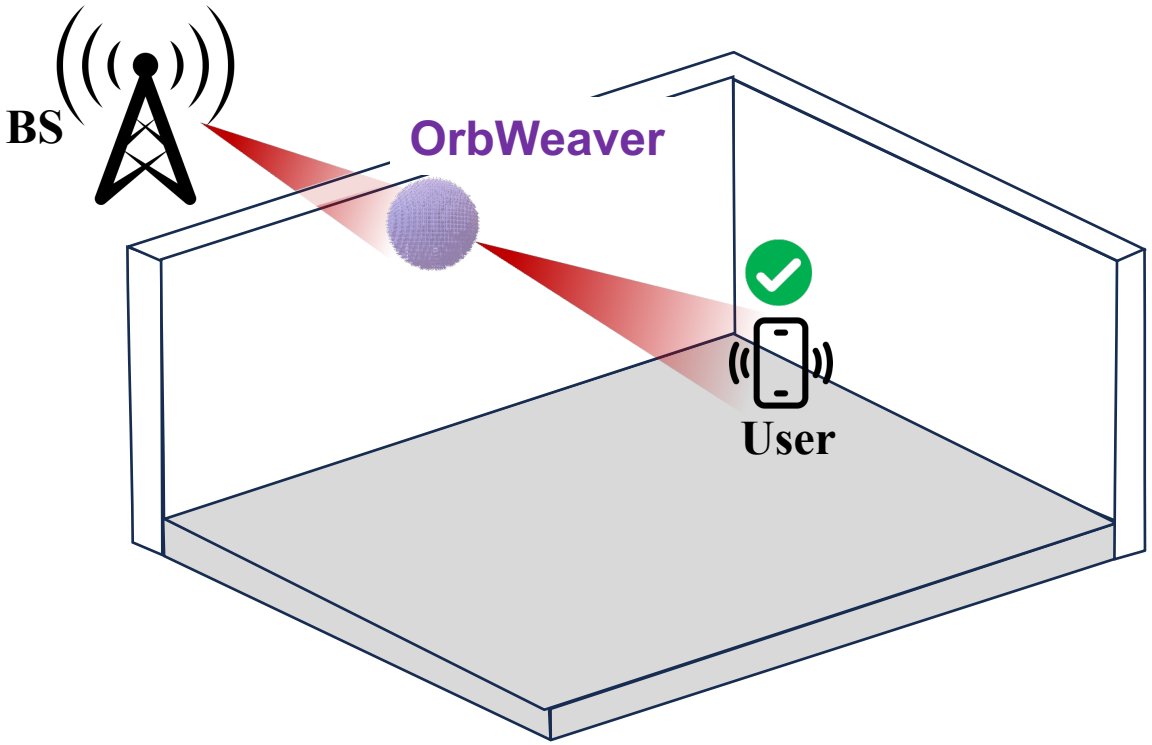


Can we simply reroute the beam in space,
instead of programming thousands of meta-atoms?

Our Answer： OrbWeaver — A Geometric Optics-Based RIS for Omnidirectional 3D Beam Steering

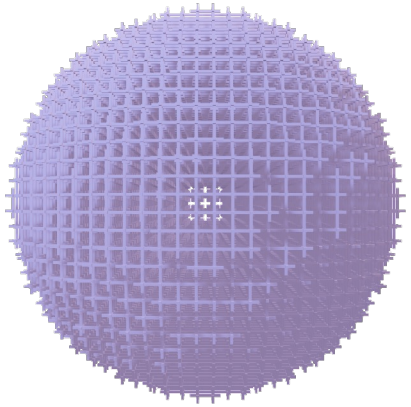


Reflective Mode

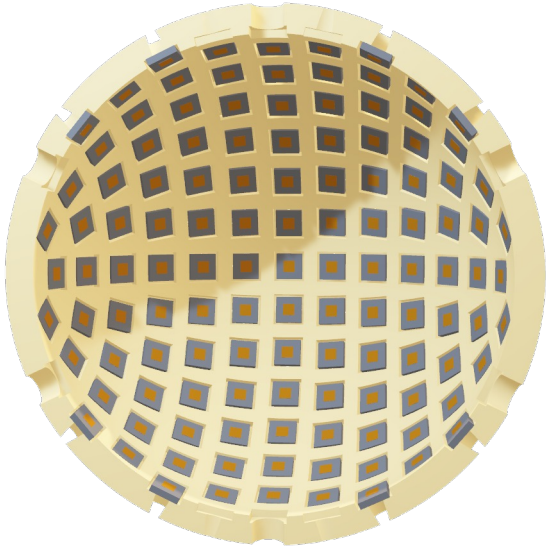


Transmissive Mode

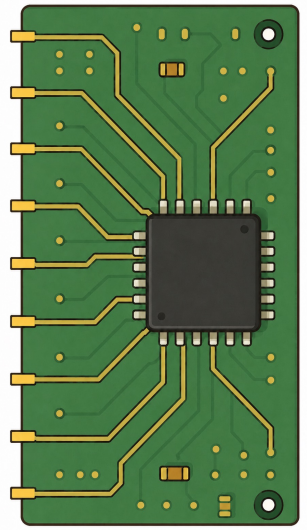
OrbWeaver Architecture: Three Core Components



3D-Printed Luneburg Lens

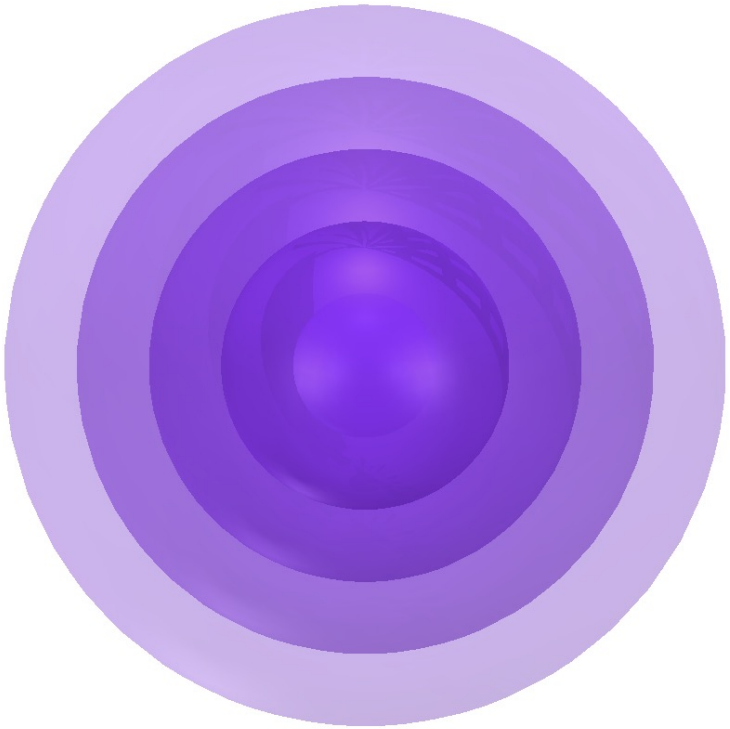


Hemispherical Antenna Array

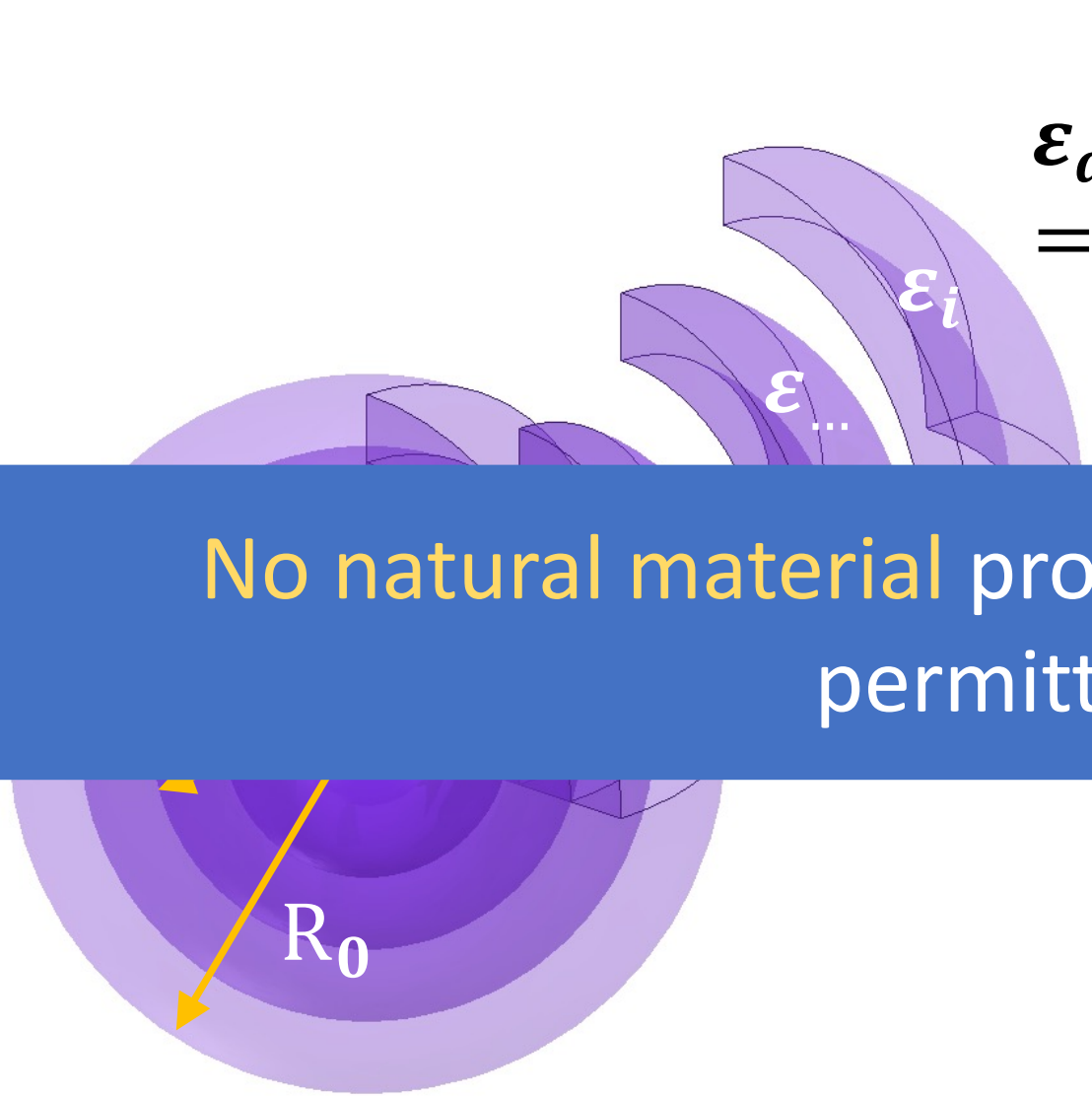


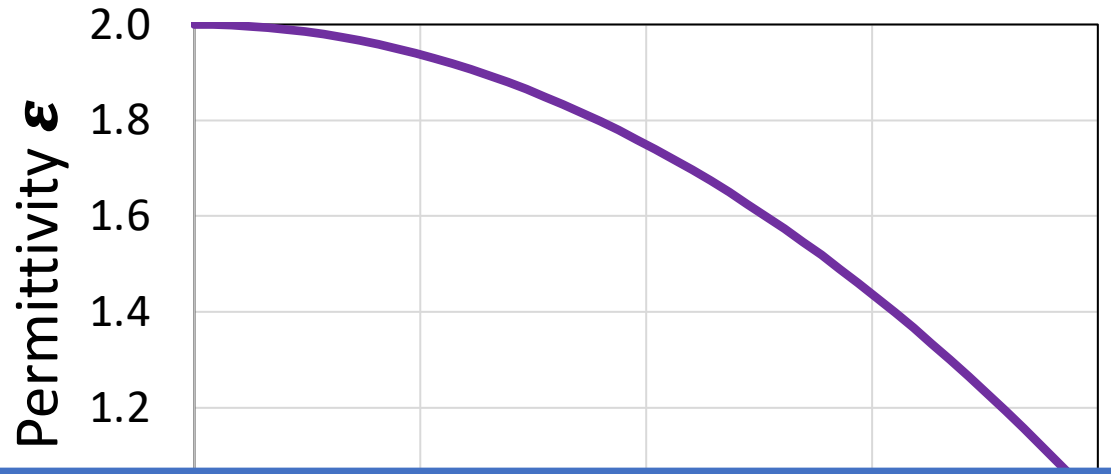
RF Switching Fabric

A gradient-index lens bends waves through its internal structure

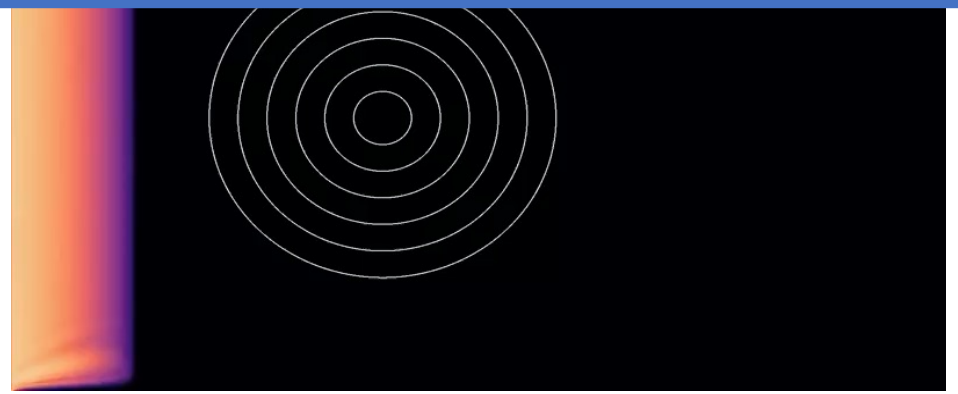


A gradient-index lens bends waves through its internal structure

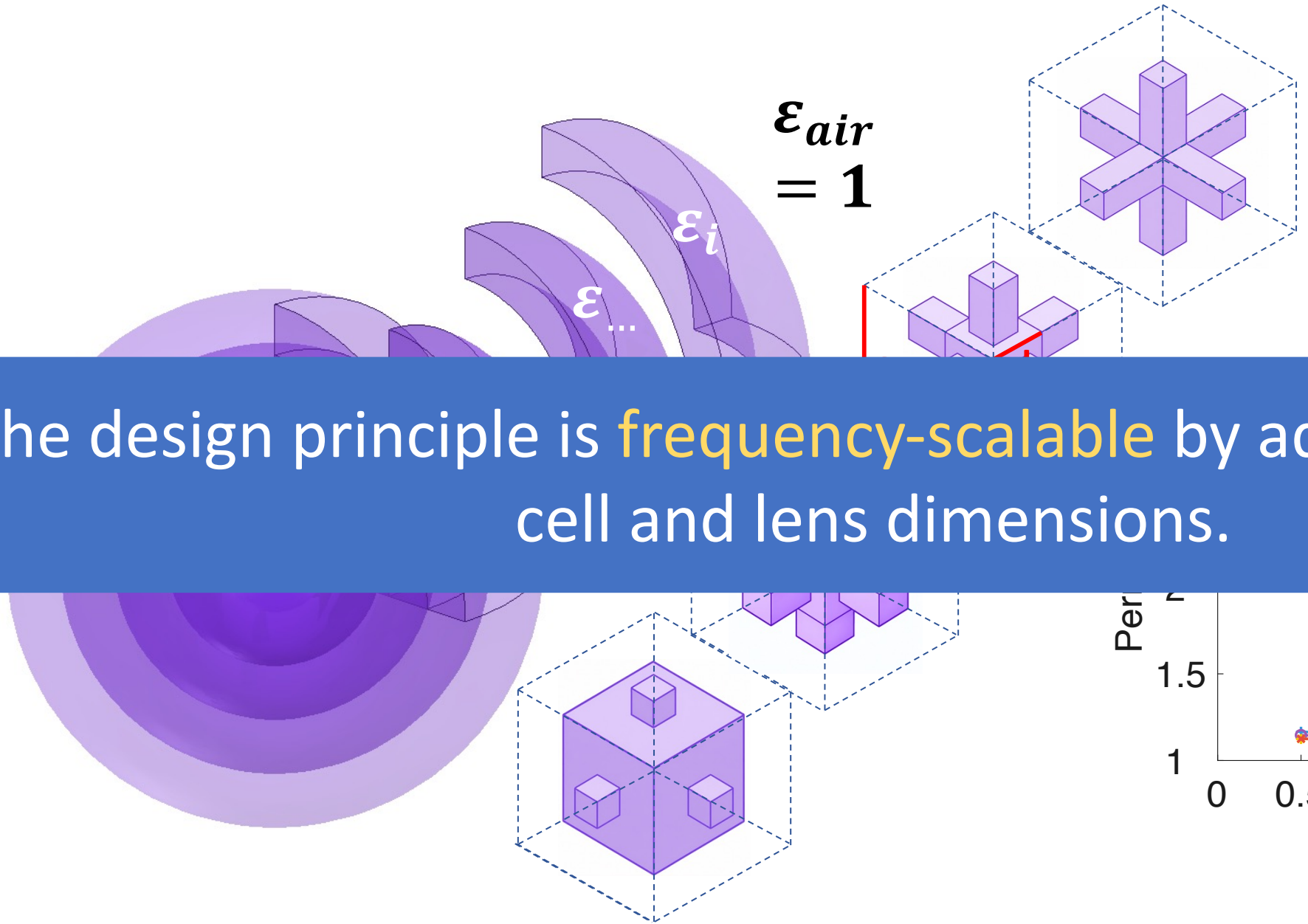




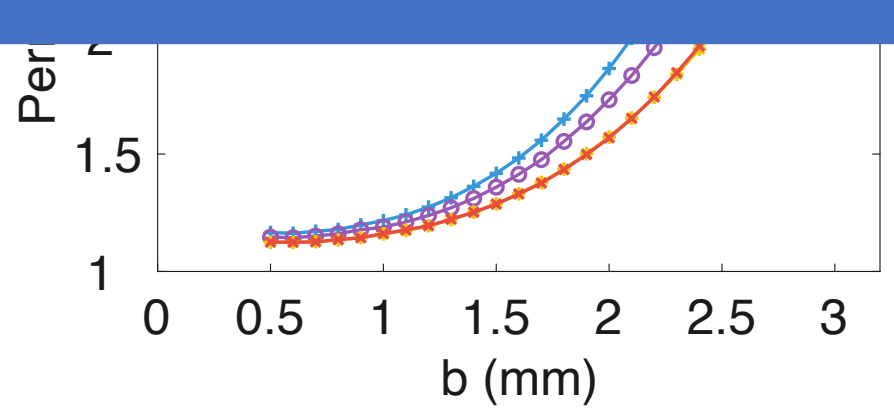
No natural material provides the required continuous permittivity gradient.



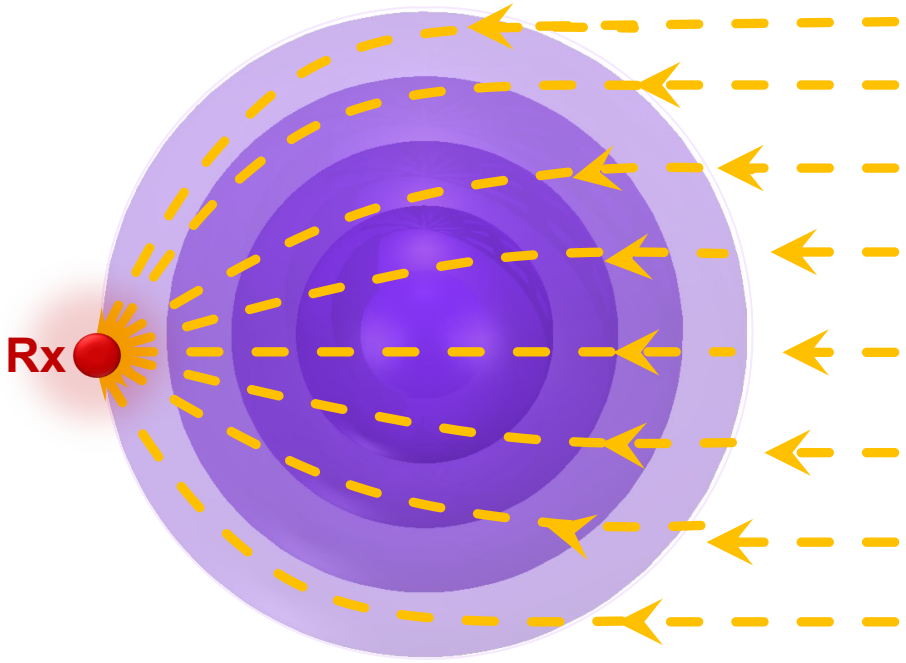
3D Printing Approximates the Gradient Using Tunable Unit Cells



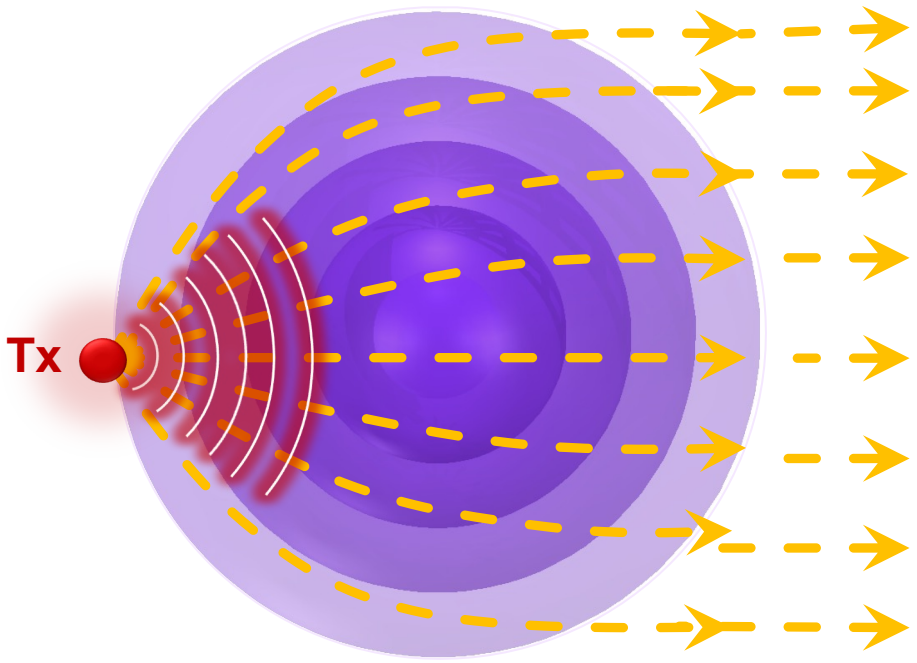
The design principle is frequency-scalable by adjusting the unit-cell and lens dimensions.



The Luneburg Lens Focuses and Collimates Signals Reciprocally

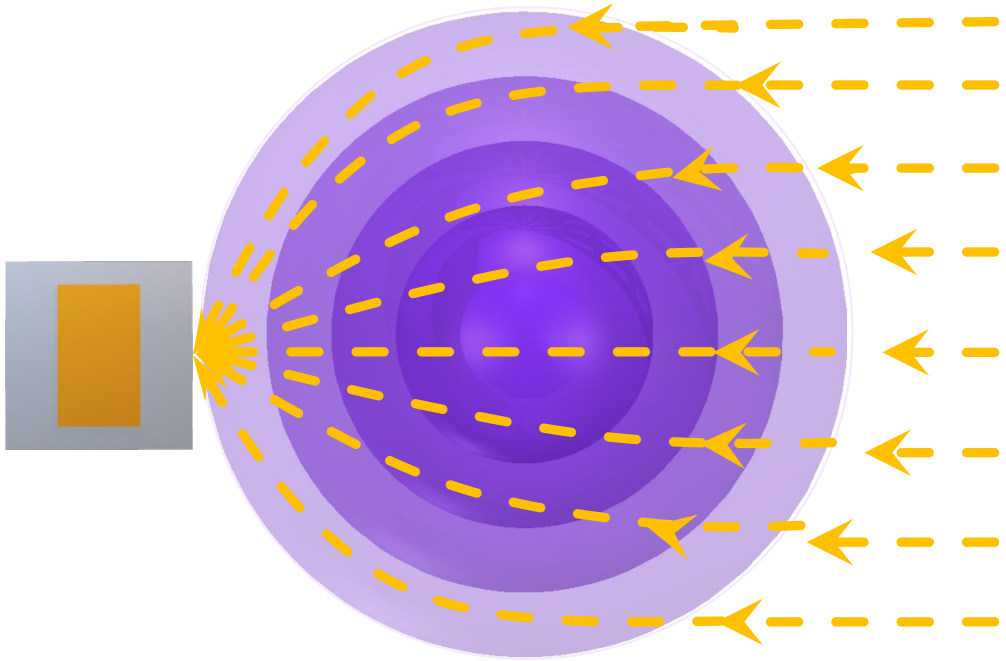


As Rx — Signal focusing

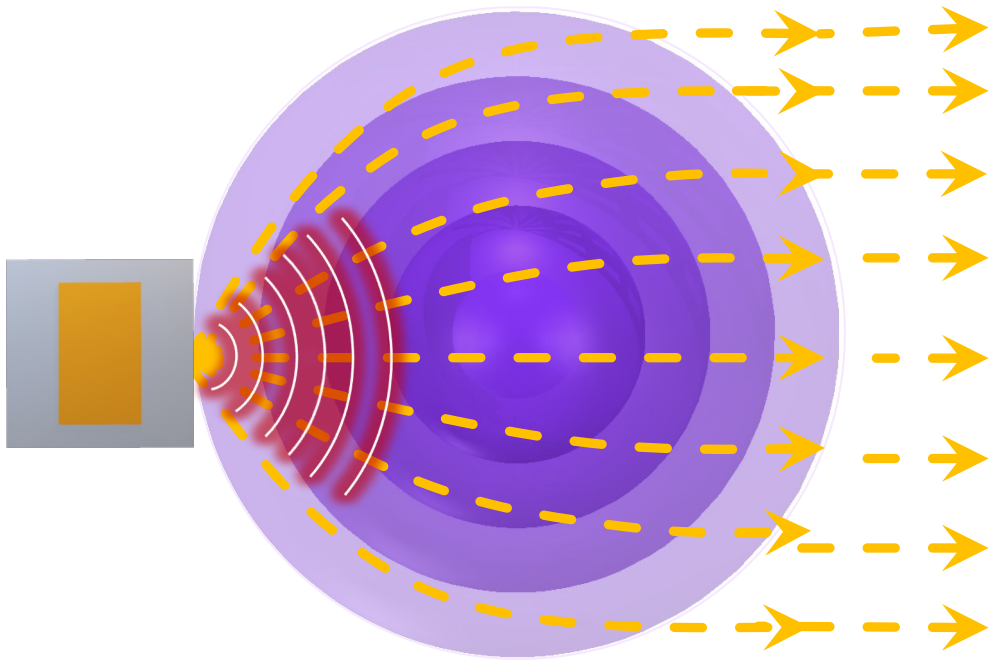


As Tx — Signal collimation

Patch antennas bridge the lens and the RF switching circuit



As Rx — Signal focusing



As Tx — Signal collimation

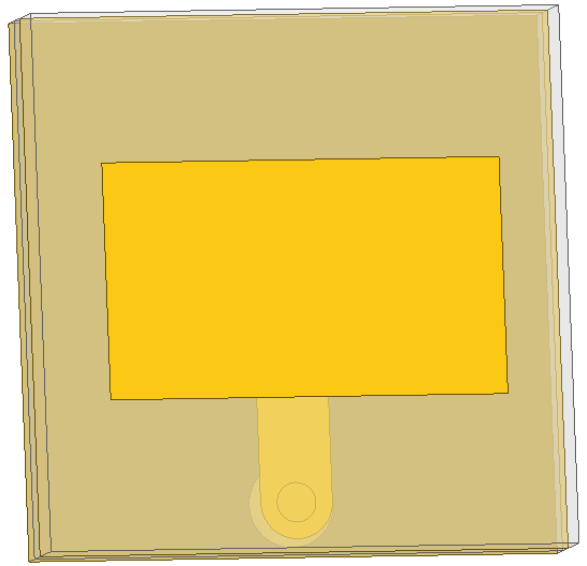
Antenna design balances bandwidth, gain, size, and angular resolution



Wide operating bandwidth

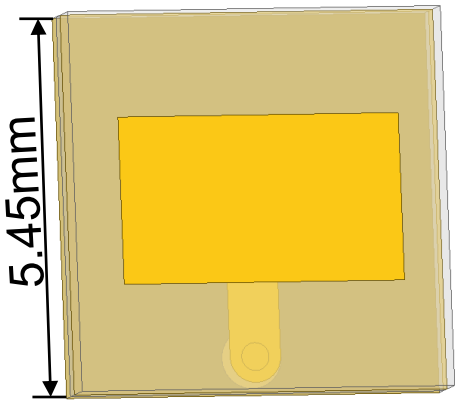
High gain with the lens

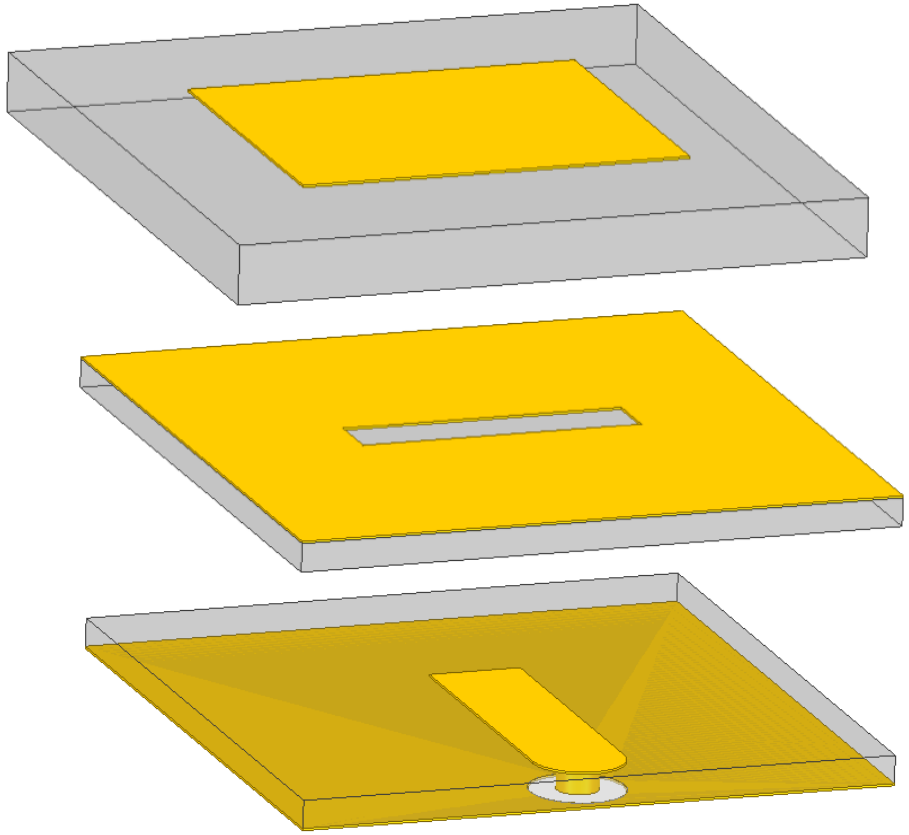
Compact element size for hemispherical integration



Multi-layer stacked microstrip antenna

Antenna design balances bandwidth, gain, size, and angular resolution





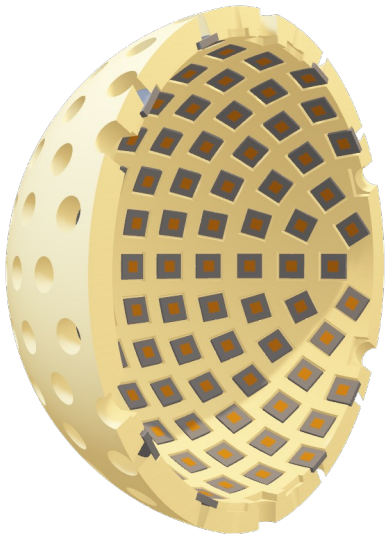
Primary radiation unit

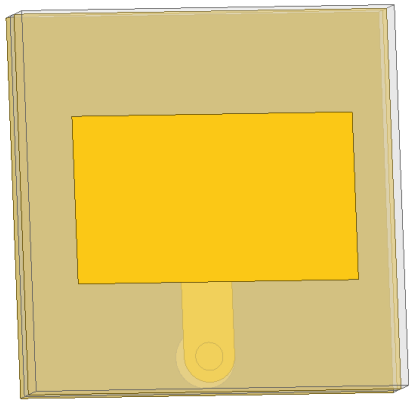
Shared GND with thin strip slot

Feeding network layer

Our patch antenna supports a wideband lens–antenna response, delivering over 23 dBi gain across the n257 band(26.5–29.5 GHz) and a 25.8 dBi peak gain.

Antenna design balances bandwidth, gain, size, and angular resolution



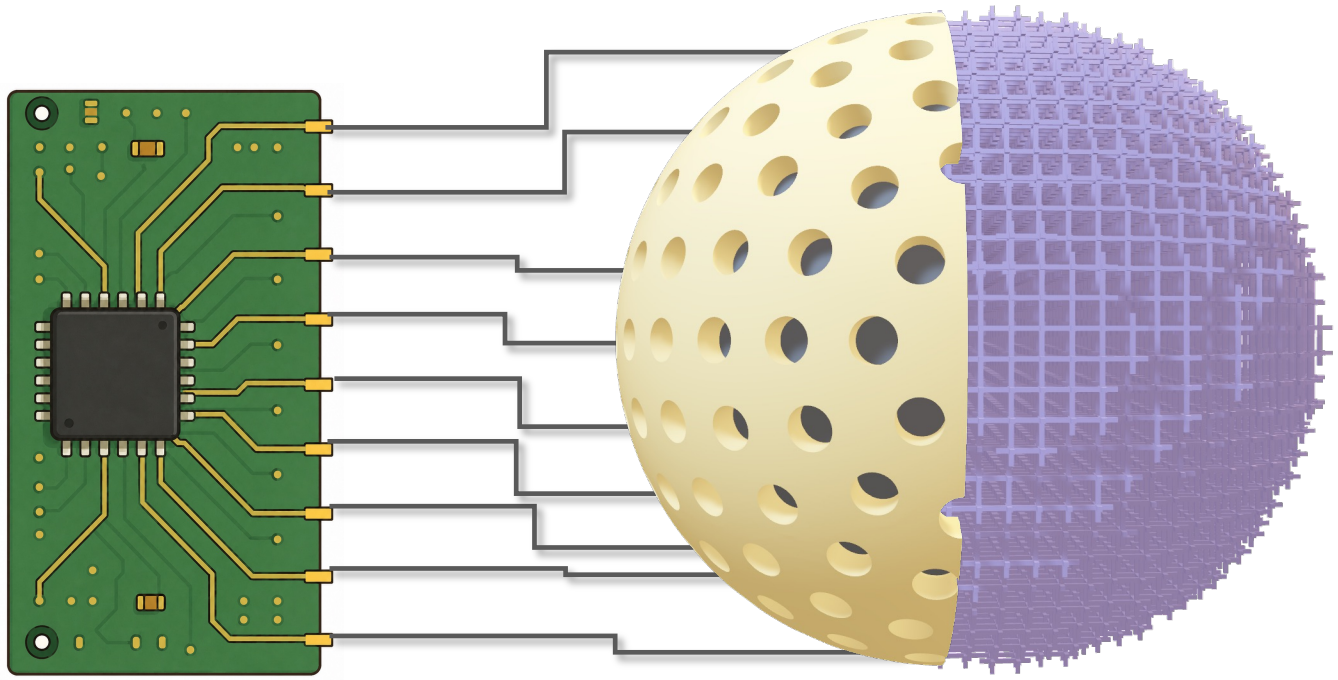


From antenna ports to programmable beam routing

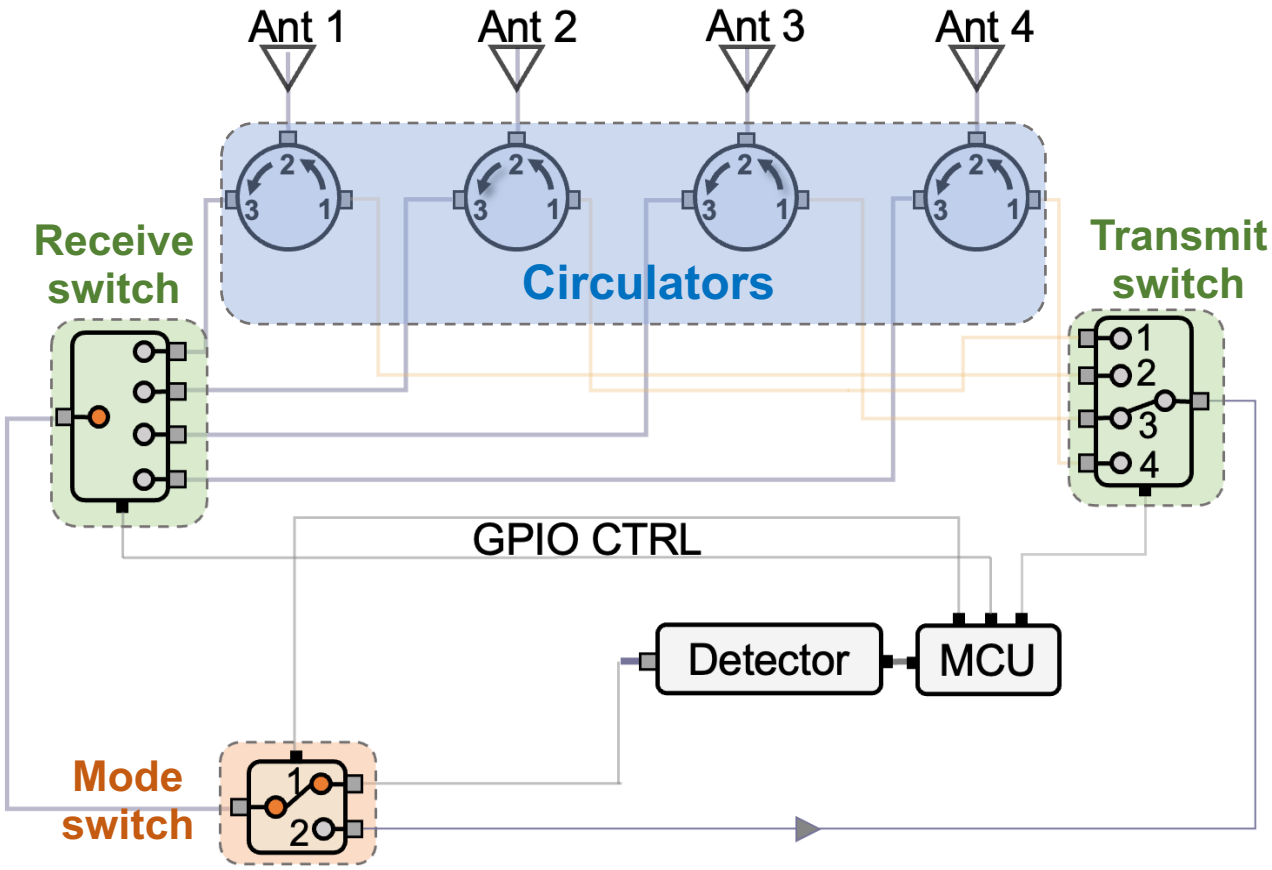


How can we route an incoming beam to any desired direction?

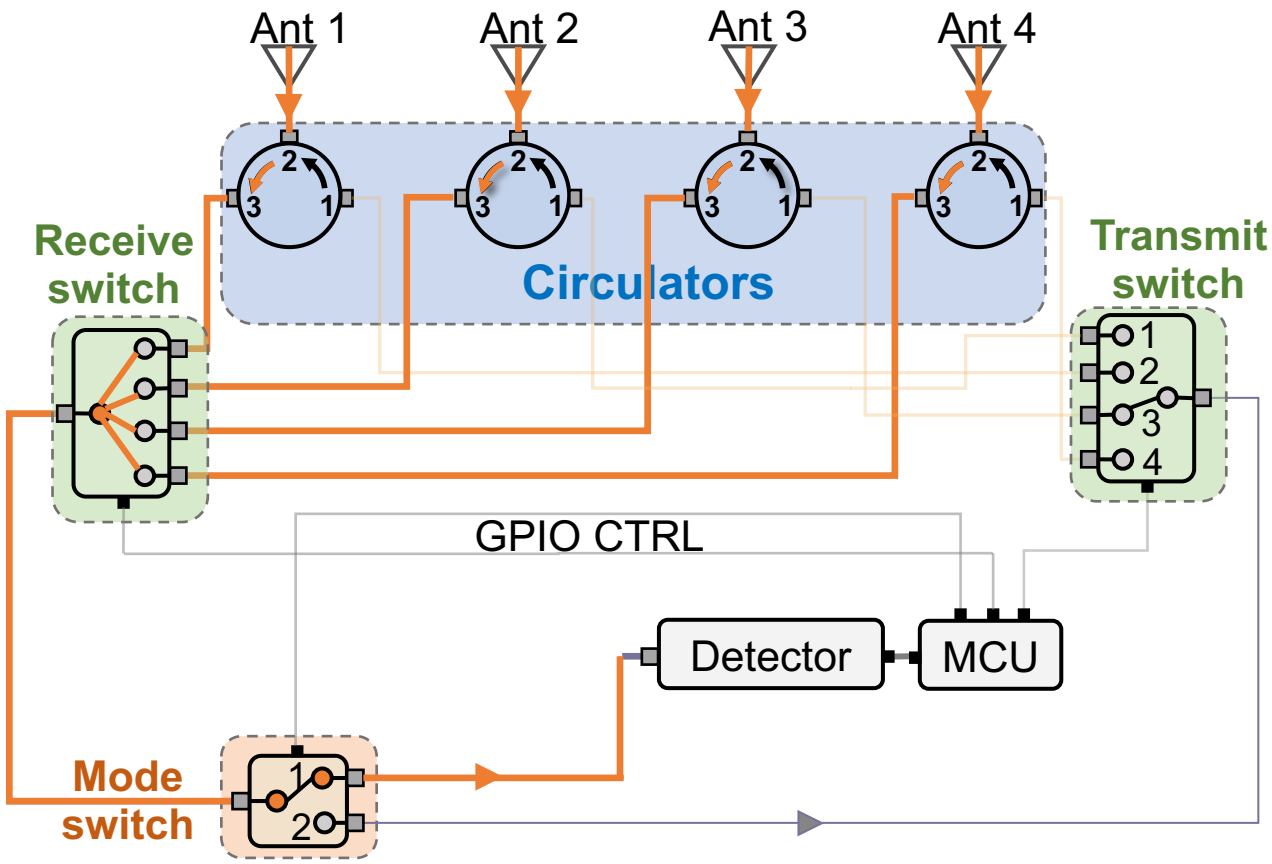
From antenna ports to programmable beam routing



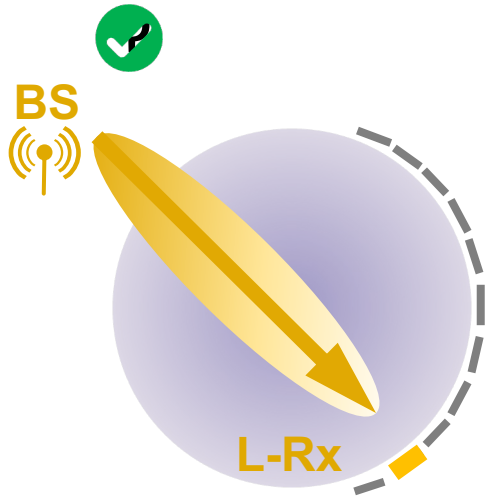
Reconfigurable RF Switching Fabric



Reconfigurable RF Switching Fabric

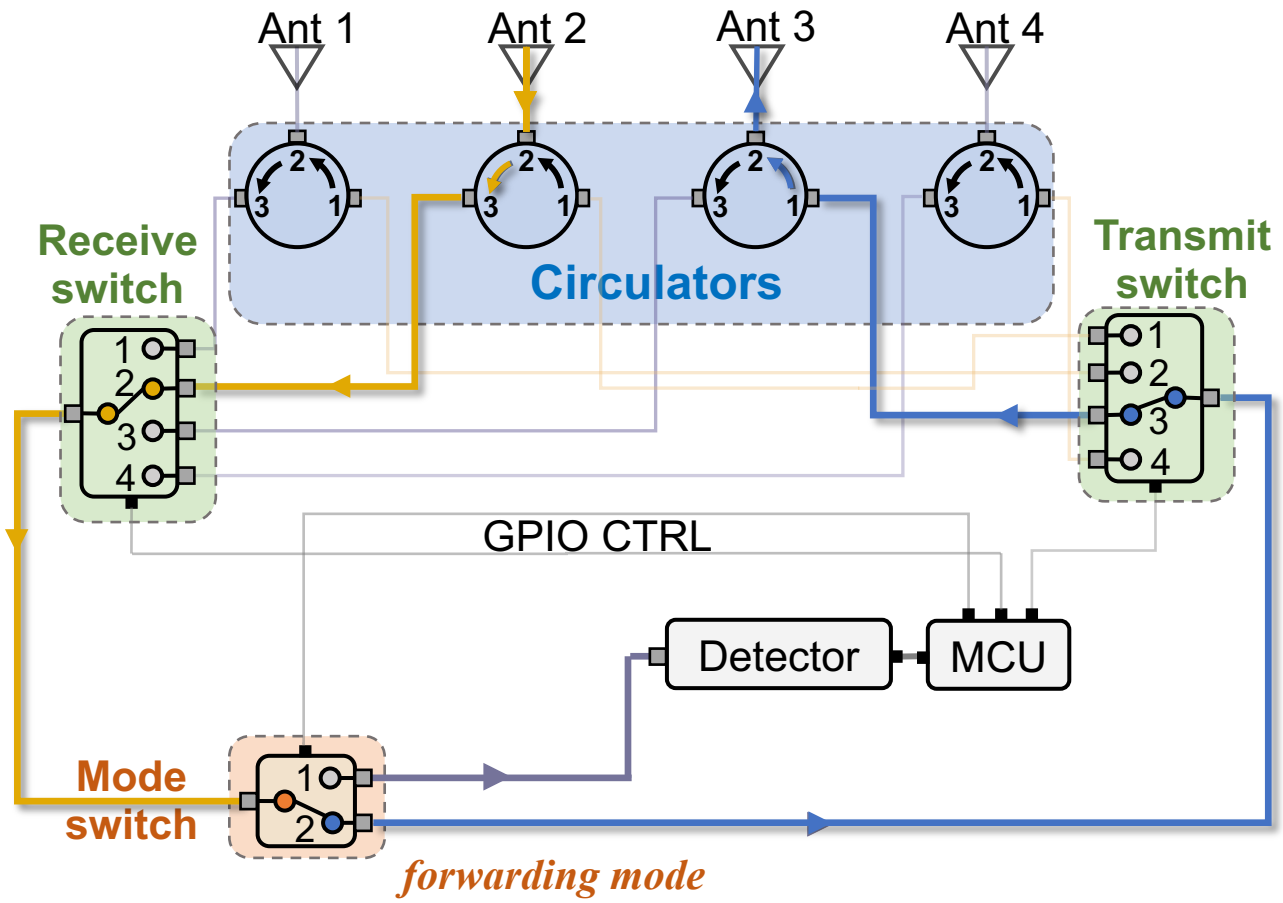


Sensing mode

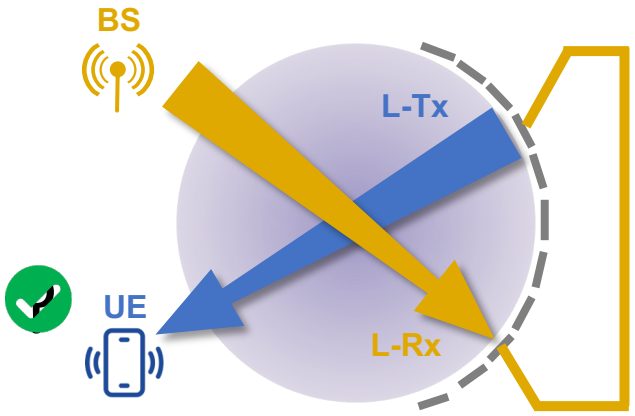


The strongest antenna port directly reveals the incident direction.

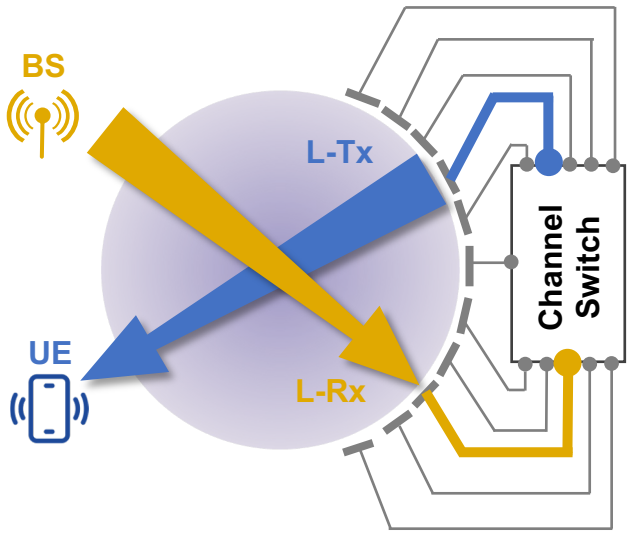
Reconfigurable RF Switching Fabric



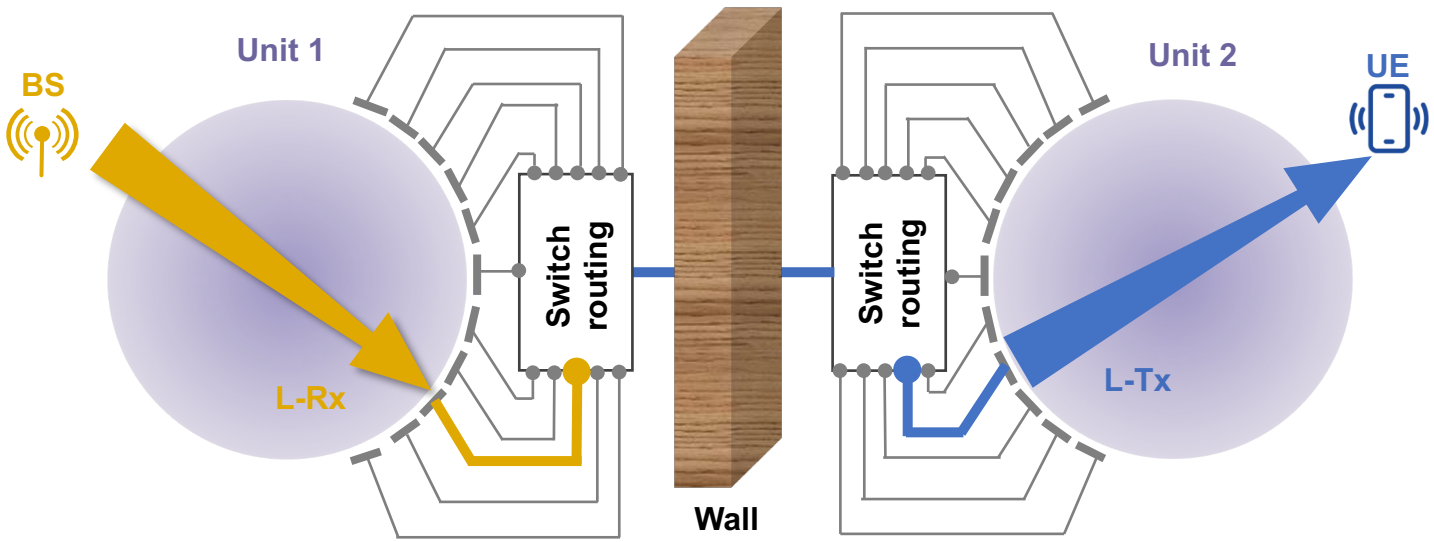
Forwarding mode



One fabric, multiple capabilities

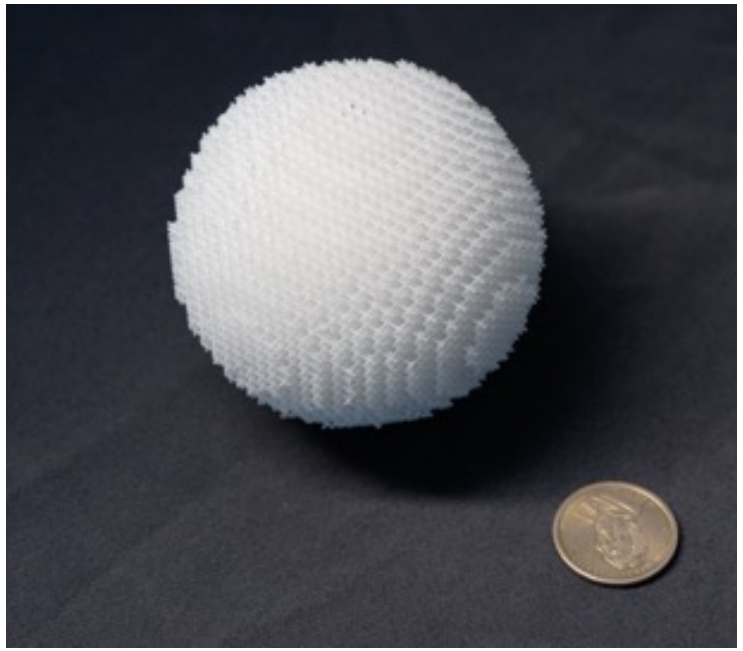


Reflective Mode

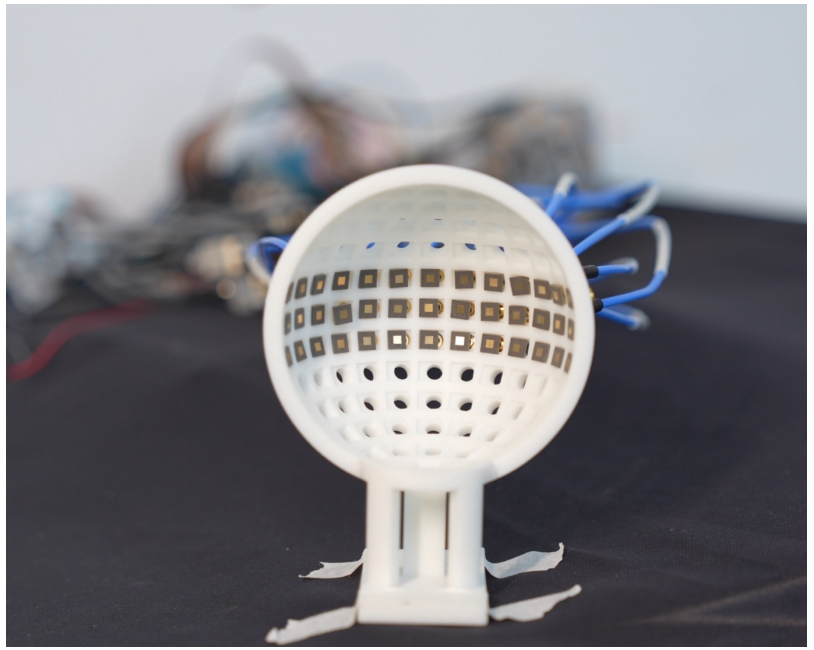


Transmissive Mode

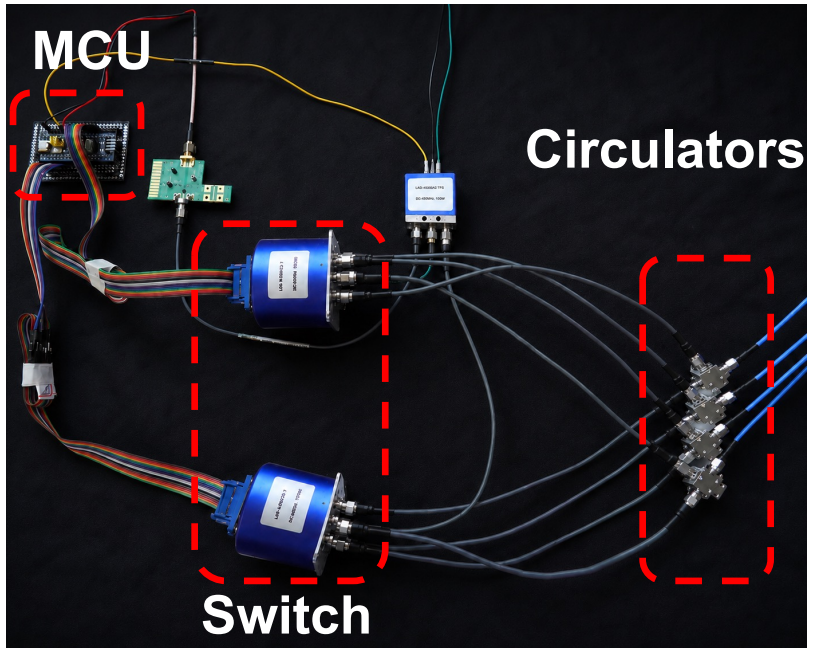
OrbWeaver Components



Luneburg Lens



Antenna Array



Switching Circuit


UE
BS

Conclusion

OrbWeaver: a geometric-optics-based RIS for next-generation wireless networks

- **Simple hardware and control:** 3D-printed lens + RF switching fabric.
- **Wide-angle 3D coverage:** stable steering from −85° to +85°.
- **Built-in sensing for efficient alignment:** efficient beam alignment without prior location information.
- **Frequency-scalable architecture:** validated at mmWave and sub-6 GHz.



OrbWeaver: A Geometric Optics-Based RIS for Omnidirectional 3D Beam Steering

Q & A

Thank You!